Internship Roadmap

Duration: 20 January – 20 July
Intern: Aman Srivastav
Domain: Embedded Systems & Intelligent Automation Platforms

1.Summary

This internship focuses on structured engineering contributions across embedded systems, sensor calibration, compliance validation systems, and AI-driven automation.

The work spans six major phases:

1. Embedded UI Stabilization
2. Moisture Calibration Modeling
3. Temperature Compensation Engineering
4. Food Labz Digital Compliance Platform
5. Computer Vision-Based Food Waste Detection
6. NIR Spectral Automation & Integration

2. Detailed Breakdown

Track 1 – Embedded Systems Infrastructure (January)

Objective

Establish a stable Wayland-based UI system for headless Raspberry Pi deployment.

Work Completed

- Raspberry Pi OS Lite setup
- Wayland text-input validation
- Virtual keyboard implementation

Outcome

Stable embedded UI stack for production-style deployment.

Track 2 – Moisture Calibration & Temperature Compensation (February – April)

Phase 2A – Research & Calibration (February)

Research Contribution

- Assisted in research of NIR sensors and quality meter systems
- Studied wavelength vs intensity signal behavior
- Supported experimental data understanding

Calibration Engineering

- Linear regression modeling
- Piecewise linear calibration
- Continuity constraint enforcement
- RMSE & MAE optimization
- Residual analysis

Phase 2B – Temperature Compensation (March–April)

Objective

Improve model reliability under varying temperature conditions.

Work

- Moisture = f(Signal, Temperature) modeling
- Drift coefficient estimation
- Compensation equation development
- Multi-temperature validation
- Stability benchmarking

Outcome

Temperature-compensated moisture calibration model.

Track 3 – Food Labz Digital Compliance Platform (February – July)

Phase 3A – Architecture & Army Use Case (February–April)

Scope

Development of digital report validation system focused initially on Army use case.

Work Completed

- Backend architecture design (Django)
- PDF upload API
- SHA-256 hashing for duplicate detection
- OCR pipeline (PDF → Image → Text)
- Structured certificate parsing
- Validation engine logic

  Duplicate ULR detection
  Certificate existence check
  Expiry validation
  Issue date validation

Outcome

Functional Army-focused digital certificate verification module.

Phase 3B – Core Validation Platform Expansion (April–May)

Work

- Validation logic refinement
- Composite uniqueness enforcement
- Master dataset cleaning
- Index optimization
- Structured API response design

Outcome

Scalable backend validation engine.

Phase 3C – FSSAI Integration & Enterprise Scaling (May–July)

Scope

Expansion into a full compliance platform integrating regulatory datasets.

Work

- FSSAI dataset integration
- Data normalization & cleaning
- Large dataset handling (100k+ records target)
- PostgreSQL migration planning
- Role-based authentication
- Audit logging
- Dashboard and analytics preparation
- Deployment planning (cloud readiness)

Outcome

Enterprise-ready food compliance verification system.

Track 4 – Computer Vision: Food Waste Detection (March)

Objective

Develop AI-based system for food waste classification.

Work

- Dataset collection & annotation
- Image preprocessing
- CNN training
- Accuracy benchmarking
- Integration into reporting pipeline

Outcome

Prototype food waste detection model.

Track 5 – NIR Spectral Automation & System Integration (July)

Objective

Automate spectral data processing and integrate complete system pipeline.

Work

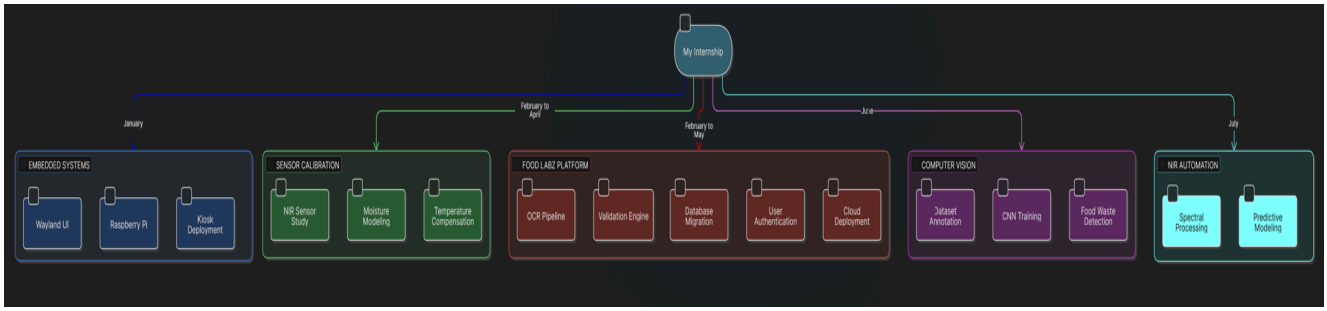
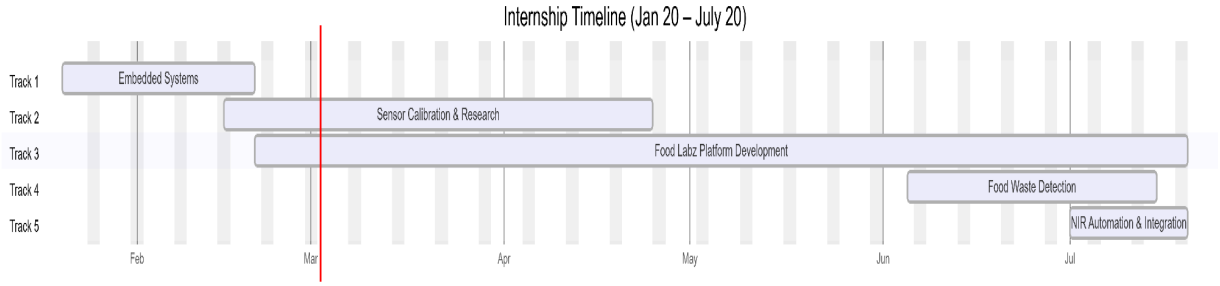
- Spectral file parsing
- Signal smoothing
- Feature extraction
- Predictive modeling pipeline
- End-to-end integration
- Final validation & documentation

Outcome

Automated spectral-to-prediction system.

3. Visual Roadmap Overview

3.1 Mind Map



3.2 Project Timeline – Gantt Chart